

Data Tasks: Collection, Cleaning & Analysis

Time-Aware Explainable ML for Early At-Risk Student Prediction on OULAD

Group 1 – DSP391m

FPT University

Data Science Capstone (DSP391m) ■ Report 2

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Data Requirements → OULAD Sources

Requirement	Why required	OULAD source
Binary at-risk label	Prediction target (all RQs)	<code>studentInfo.final_result</code>
Time-stamped engagement	Checkpoint features (RQ1/2)	<code>studentVle (~10.6M)</code> , <code>vle</code>
Academic performance	Early score signals	<code>studentAssessment</code> , <code>assessments</code>
Demographic context	Covariates + fairness	<code>studentInfo</code> , <code>studentRegistration</code>
Course timeline	Date → progress %	<code>courses</code> (module length)

⇒ Grain = one row per (student, module, presentation): **32,593** rows, **28,785** students, **22** module-presentations.

OULAD Composition (7 Relational Tables)

Table	Content	Rows	Time?
studentInfo	Demographics + final_result	32,593	no
studentRegistration	Registration / withdrawal dates	32,593	part
studentVle	Daily clickstream interactions	10,655,280	yes
studentAssessment	Submission dates + scores	~173,912	yes
assessments	Type, weight, deadline	~206	part
vle	Catalogue of activity types	~6,364	no
courses	Module length (days)	22	no

⇒ The two date columns – *studentVle.date*, *studentAssessment.date_submitted* – enable every time-aware cut.

Working at Scale – Understanding 10.6M Rows with ClickHouse

The problem – you cannot eyeball it

- studentVle = 10,655,280 rows / 433 MB: too big for Excel; reloading it in pandas *per question* costs GBs of RAM.
- **Method:** clickhouse-local – one binary, no server, runs SQL straight on CSV/Parquet. Data stays on disk; we push the query down.

The type-inference trap

- Quoted CSV $\Rightarrow$ every column read as String; `max(date)` returns "99" (string compare, "99">"269").
- Declare the schema $\Rightarrow$ true range date -25 $\rightarrow$ 269.

$\Rightarrow$ Push SQL to the data: 10.6M rows understood in <1s, and **every number on these slides is reproducible** (notebooks/00_data_understanding).

	Size	count()
CSV	433 MB	≈ 14.6 s
Parquet	22.9 MB	≈ 0.55 s
Gain	$\sim 20\times$	$\sim 26\times$

Profiled in one pass

- 26,074 students actually click (of 28,785).
- per-event clicks: median 2, skew 122.
- total_clicks skew 3.0 $\Rightarrow$ motivates log1p.

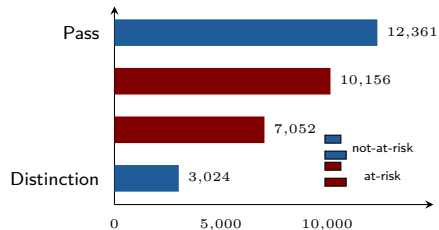
Target Variable & Class Balance

final_result	Count	Label
Pass	12,361	not-at-risk (0)
Distinction	3,024	not-at-risk (0)
Withdrawn	10,156	at-risk (1)
Fail	7,052	at-risk (1)

`at_risk = {Fail, Withdrawn}`

At-risk **52.8%** ■ imbalance ratio **1.12** (mild).

Metric focus: **PR-AUC** + **recall** on at-risk.



Withdrawn is the largest single class.

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Choosing a Data Source

Criterion	Internal DB	API	Scraping	Public
Immediate availability	Low	Medium	Medium	High
Already anonymised	No	No	No	Yes
Reproducibility	Low	Low	Very low	High
Comparability w/ prior work	Low	Low	Low	High
Ethical / legal complexity	High	Medium	High	Low (CC)

⇒ *OULAD is **public secondary data**: we trade collection-design control for reproducibility & comparability with the base studies (Adnan 2021; Tomasevic 2020).*

Why OULAD – and License & Ethics

Fit to the project

- Publicly downloadable – reproducible by anyone.
- All 3 feature groups: demographics, VLE, assessments.
- Labelled `final_result` → `at_risk`.
- Same dataset as the base studies ⇒ direct comparison.
- Static snapshot – identical file for everyone.

License & ethics

- **CC-BY 4.0** – reuse with attribution only.
- Anonymised at source: no IDs; region-level; banded age & IMD.
- No re-identification; no external linkage.
- Raw CSVs read-only + md5 verification.
- Cite Kuzilek et al. (2017), *Scientific Data*.

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Data Quality Profile – Where Are the Defects?

Column	#Missing	%	Mechanism	Handling (where)
date_unregistration	22,521	69.1	structural	dropped – not a feature
imd_band	1,111	3.41	MAR / MCAR	→ "Unknown" (ordinal rank 0)
date_registration	45	0.14	MCAR	→ train median
score*_to_date (3)	by design	–	MNAR	→ 0 + not_submitted flag
all 29 other columns	0	0.0	–	none needed

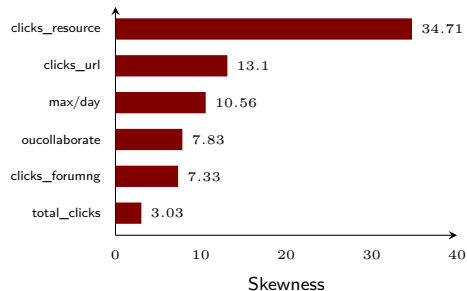
⇒ *Duplicates on the composite key: 0. After imputation: 0 NaN in every feature column (train & test).*

Missing Values – Method & Justification

Variable	Strategy	Why this, not deletion/mean
imd_band	→ "Unknown" category	Keeps the ordinal scale; never fabricates a deprivation value.
score*_to_date	→ 0 + not_submitted	A missing score <i>is</i> a signal (nothing handed in yet); the flag stops 0 being read as a real zero score.
date_registration	→ train median	Only 45 rows, MCAR; train-only fit avoids leakage.
date_unregistration	dropped	69.1% missing is structural; imputing a fake date has no meaning.

⇒ *All statistics are learned on the **training fold only**, then applied to test (no leakage).*

Outliers – Severe Right-Skew in the Clickstream



Evidence (IQR rule, train)

- clicks_resource: skew 34.7, kurtosis 2125, max 5,147 vs median 19.
- max_clicks_single_day: max 6,988 vs median 74.
- A symmetric variable has skew ≈ 0 ; here it is huge
⇒ long right tail.

⇒ *Untreated, these extremes dominate gradients/distances and inflate variance.*

Outliers – Method, Choice & Evaluation

Strategy	Applied to	Why / evaluation
log1p	all clickstream counts (total_clicks, n_days_active, 8×clicks_*, max_clicks_single_day)	Compresses the long tail, maps 0→0 (many zero-activity students).
winsorise 1%	studied_credits, num_of_prev_attempts, weighted_score, days_since_last_activity	Clips top/bottom 1%, keeps the rank; safer than log on a score.
none	mean_score_to_date, n_assessments, date_registration	IQR upper bound 103 > max 100 ⇒ no true outlier.

⇒ ***No row is ever deleted*** – only transformed; thresholds learned per training fold.

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Variable Typing – 28 Features, 5 Types

Type	#	Examples	Encoder / scaler
Numeric	19	clicks, scores, studied_credits	StandardScaler
Ordinal	3	highest_education, imd_band, age_band	OrdinalEncoder
Nominal	3	region, code_module, code_presentation	OneHotEncoder
Binary	2	gender, disability	BinaryEncoder
Indicator	1	not_submitted	passthrough

⇒ *Type drives the transformer: wrong type = lost rank (ordinal) or a false order (nominal).*

Encoding – Method & Rationale

Encoder	Key configuration & why
OrdinalEncoder	Explicit category order (e.g. No Formal < ... < Post Grad); unknown=-1. Preserves rank, no dimensionality blow-up.
OneHotEncoder	handle_unknown=ignore, drop=None. No false order; every category stays visible for SHAP/LIME.
BinaryEncoder	Fixed map M/Y→1, F/N→0. Deterministic, no fitting needed.
passthrough	not_submitted already 0/1 from feature engineering.

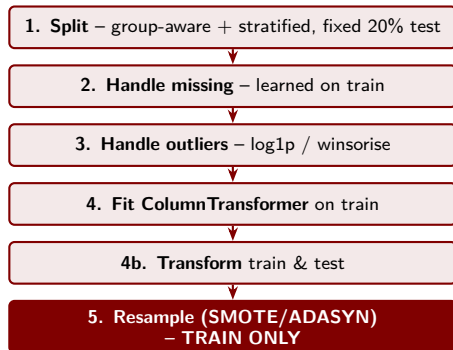
⇒ *drop=None is deliberate: dropping a reference column would hide that category from explanations.*

Standardisation: Before → After

Component	Before	After
Labels	class names	binary at-risk = 1 / 0
Categorical	text values	numeric (ordinal/one-hot/binary)
Numeric	raw, skewed, 0–24k	log1p/winsorise + z-score (mean 0, sd 1)
Matrix	28 mixed-scale cols	49-column dense, 0 NaN
Names	raw	snake_case, transformer-prefixed

⇒ *Scaler fitted on **train only** (`scaler.mean_` from train); test uses `.transform()` – never `.fit_transform()`.*

Preprocessing Sequence – Fit on Train Only



Anti-leakage reasons

- Split **first** – no statistic flows test→train.
- Impute/encode/scale **fit on train**, apply to test.
- Outlier thresholds per fold; **no rows dropped**.
- Resample **train only** – test keeps real class ratio.
- Mirrors the time cut (cut_at_checkpoint).

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Why the Split Needs Care

1. Grouped

A student spans several module-presentations $\Rightarrow$ split by `id_student`, else group leakage.

2. Imbalanced

At-risk $\approx 52.8\%$ $\Rightarrow$ *stratify* so the test ratio cannot drift.

3. Time-aware

6 checkpoints share one axis $\Rightarrow$ test set *identical* at each.

Train rows	Test rows	Test students	At-risk tr/te	Overlap
26,104	6,489	5,756	.530 / .520	0

$\Rightarrow$ *Verified on master_raw (32,593 rows), identical across all six checkpoints.*

Splitting Strategies Compared

Strategy	Pros	Cons
Single hold-out	simple, fast	high variance; one partition
k-fold CV	uses all data; lower variance	$k \times$ cost; still seed-dependent
Repeated k-fold	robust mean$\pm$std; seed-independent	highest cost
Nested CV	unbiased with tuning	very expensive, complex

⇒ **Chosen:** fixed 20% hold-out + **5-fold** $\times$ **5-seed** repeated CV (*StratifiedGroupKFold*) – robust yet comparable across checkpoints. Report mean $\pm$ std PR-AUC / recall.

Leakage Prevention – and It Is Tested

Three temporal rules (per checkpoint t , $\text{cutoff} = \text{round}(\text{length} \times t/100)$)

- 1 Drop assessment submissions with `date_submitted > cutoff`.
- 2 Drop VLE interactions with `date > cutoff (cut_at_checkpoint)`.
- 3 Withdrawn-before- t kept & labelled at-risk – low activity is signal, not leakage.

+ **Fit-on-train principle** for every learner (imputers, encoders, scaler, resampler).

Verified automatically by `tests/test_leakage.py`

19 / 19 tests pass

no record beyond cutoff ▪ counts non-decreasing in t ▪ $t=100\%$ keeps all ▪ 0 overlap ▪ ratio preserved ▪ train-only impute/winsorise ▪
no-leak feature allow-list

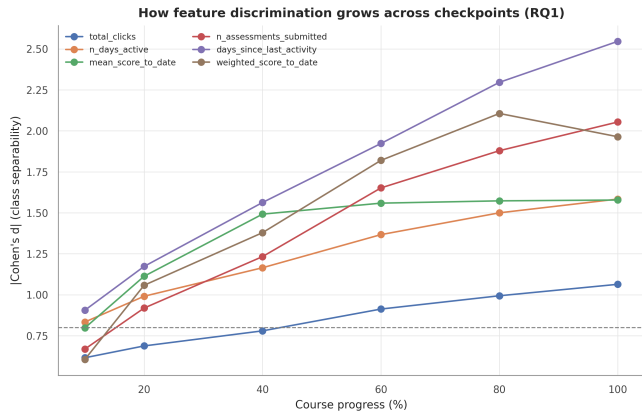
Time-Aware Checkpoints – Signal Grows With Time

Six checkpoints: 10 / 20 / 40 / 60 / 80 / 100%.

Earliest checkpoint reaching $d \geq 0.8$:

Feature	at $t =$
n_days_active, days_idle	10%
mean_score / n_assess / weighted_score	20%
total_clicks	60%

⇒ Useful signal exists **early** – the basis for early intervention.

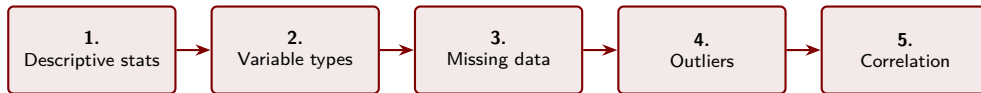


Project figure: reports/figures/time_discrimination_curve.png

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EDA Roadmap



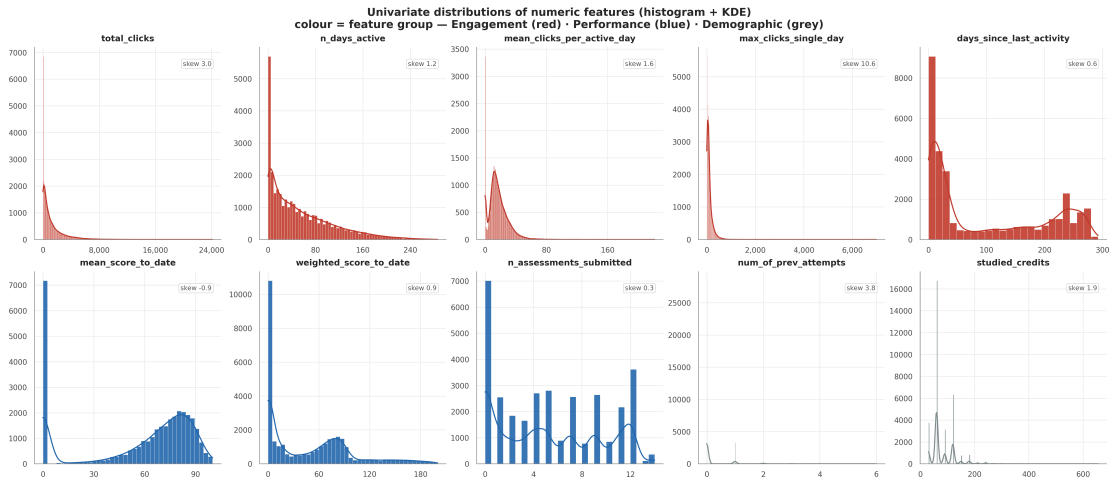
We explore at three levels: **univariate** (one variable) → **bivariate** (vs the target) → **multivariate** (between variables).
Raw data → patterns → conclusions that guide modelling.

Descriptive Statistics (Univariate Numeric)

Feature	Mean	Median	Std	Min	Max	Skew
total_clicks	1215.1	602.0	1692.6	0	24,139	3.03
n_days_active	55.5	40.0	54.5	0	286	1.18
days_since_last_activity	97.0	34.0	100.6	0	292	0.57
mean_score_to_date	57.0	70.2	33.3	0	100	-0.85
weighted_score_to_date	48.0	39.5	49.3	0	200	0.86
n_assessments_submitted	5.3	5.0	4.3	0	14	0.28
studied_credits	79.8	60.0	41.1	30	655	1.88
max_clicks_single_day	109.9	74.0	142.9	0	6,988	10.56

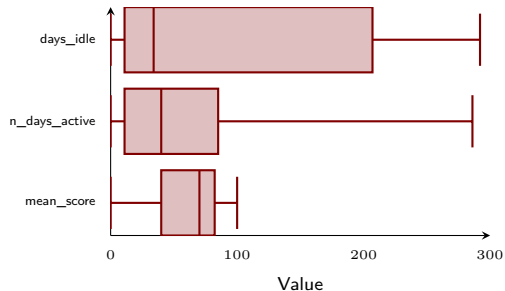
⇒ Mean $\gg$ median for clickstream (total_clicks 1215 vs 602) ⇒ *right-skew*; for mean_score median $>$ mean ⇒ *left-skew*.
Std $\approx$ or $>$ mean ⇒ *high dispersion*.

Univariate: Distributions (Histogram + KDE)



Project figure reports/figures/univariate_hist_kde.png: clickstream right-skewed, scores bounded, days_idle bimodal.

Univariate: Numeric Distributions (Box Plots)

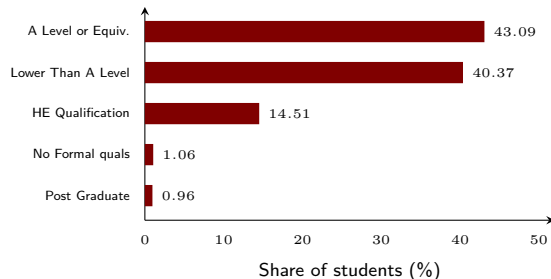


Reading the boxes

- `mean_score`: tight box, median (70) near upper quartile $\Rightarrow$ left-skew.
- `n_days_active`: wide, right-skewed spread.
- `days_idle`: huge IQR (11 $\rightarrow$ 207) $\Rightarrow$ bimodal – active vs withdrawn students.

$\Rightarrow$ Bimodal idle-time is the clearest early-warning structure in the data.

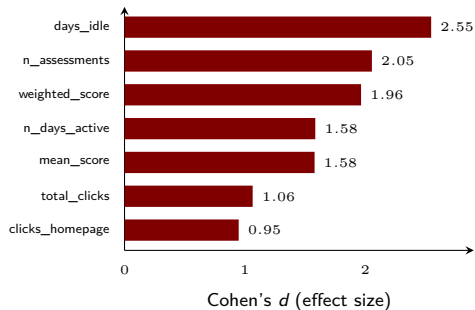
Univariate: Categorical Frequency (Highest Education)



Frequency insight

- 83% have A-Level or below.
 - Post-Graduate & No-Formal are **rare** (<1.1% each).
 - Rare levels $\Rightarrow$ one-hot with `handle_unknown=ignore`, not dropping.
- $\Rightarrow$ *Imbalanced categoricals shape the encoding choice.*

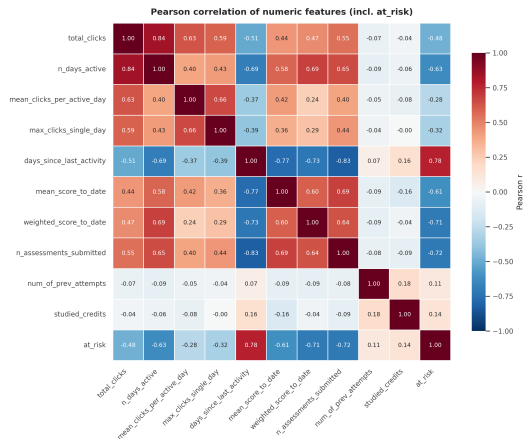
Bivariate: Numeric Features vs At-Risk (Cohen's d)



Feature	not-risk	at-risk
days_since_last_act	14.2	171.1
n_assessments	8.56	2.34
weighted_score	84.9	15.1
n_days_active	91.6	23.2
mean_score	78.5	37.8

⇒ All 19 numeric tests significant ($q < 0.05$). Behaviour, not demographics, drives risk.

Multivariate: Correlation & Multicollinearity



Top $|r|$ with at_risk

- days_idle **+0.78**; n_assessments **-0.72**;
weighted_score **-0.71**.

Multicollinear ($|r| \geq 0.8$)

- n_days_active~total_clicks: **0.84**
- days_idle~n_assessments: **-0.83**

⇒ No pair ≥ 0.95 ⇒ **no leakage**. Redundancy flagged for SHAP attribution.

Project figure: `reports/figures/corr_pearson.png`

Bivariate: Demographic Factors (Cramer's V)

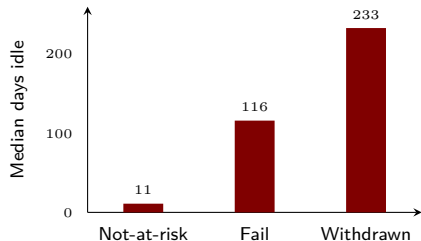
Factor	Cramer's V	χ^2 p
highest_education	0.150	$< 10^{-157}$
imd_band	0.149	$< 10^{-147}$
region	0.081	$< 10^{-38}$
age_band	0.069	$< 10^{-33}$
disability	0.059	$< 10^{-26}$
gender	0.022	5×10^{-5}

Reading it

- All significant, but associations are **weak** ($V \leq 0.15$).
- Strongest: **education & deprivation** (imd_band).
- **gender** $V=0.022$ – practically negligible.
- Cohort skew: age 0–35 **70%**; disability **9.7%**.

⇒ *Demographic effect ($V \leq 0.15$) $\ll$ behavioural effect (d up to 2.5)*
– watch fairness on imd/education, not gender.

Temporal Pattern – the “Withdrawn” Signal



Median activity by outcome

Outcome	Days idle	Total clicks
Not-at-risk	11	1,425
Fail	116	317
Withdrawn	233	89

⇒ *Withdrawn* $\approx$ near-zero engagement ⇒ trivially separable at late checkpoints. Recall/PR-AUC can look **optimistic** late – the documented Option-A limitation; the hard test is the **early** checkpoints.

EDA – Key Findings

- **Distributions:** clickstream features are strongly right-skewed (skew up to 34.7) $\Rightarrow$ log1p; scores are bounded / left-skewed.
- **Univariate:** days_idle is bimodal – the structural signature of disengagement.
- **Bivariate (numeric):** engagement & performance separate at-risk strongly (Cohen's d up to 2.55); all 19 features significant.
- **Bivariate (categorical):** demographic association is weak (Cramer's $V \leq 0.15$) – behaviour dominates.
- **Multivariate:** 2 multicollinear pairs ($|r| \geq 0.8$); none $\geq 0.95 \Rightarrow$ no leakage suspect.
- **Hypothesis for modelling:** recent inactivity + few submissions are the earliest reliable risk markers.

Summary

- 01 Required data** – 7 OULAD tables, grain (student,module,presentation); 32,593 rows, at-risk 52.8%.
- 02 Collection** – public CC-BY 4.0 secondary data; anonymised; same as base studies.
- 03 Cleaning** – 0 duplicates; missing by mechanism (MAR/MNAR/structural); skew up to 34.7 → log1p/winsorise; no row dropped.
- 04 Transformation** – 28 features, 5 types → 49-col dense matrix; fit on train only.
- 05 Split & leakage** – 20% hold-out + 5×5 CV; time-aware cuts; 19/19 tests pass.
- 06 Analysis** – behaviour dominates (days_idle $d = 2.55$); demographics weak ($V \leq 0.15$); withdrawn drives late-checkpoint optimism.

Thank you

Thank you

Key references

Kuzilek, Hlostá, Zdrahal (2017), *OULAD*, Scientific Data 4:170171 – CC-BY 4.0. Adnan et al. (2021), *IEEE Access* 9.
Tomasevic et al. (2020), *Computers & Education* 143. Chawla et al. (2002), *SMOTE*, JAIR.